

Polynomial Regression

Assignment 1 | BT2024197 | 8 October 2026

Code: https://github.com/ParthMalhotra07/ML_Assignment

1. Problem and data

This assignment uses polynomial regression for two geothermal power plant tasks. Phase 1 (var1) predicts Net Power Score from six operating settings. Phase 2 (var2) predicts Thermal Anomaly Score from three spatial coordinates. Each task has its own model, fitted to the data assigned to BT2024197.

Dataset	Inputs	Training / test rows	Maximum degree
var1	x1, x2, x3, x4, x5, x6	1,000 / 1,000	10
var2	x1, x2, x3	1,000 / 1,000	20

The training files contain the inputs and target y . The test files contain inputs only. No missing or non-finite values were found. All inputs lie in $[-1, 1]$ and are used in the supplied scale. Duplicate input rows are kept because they are separate supplied observations.

2. Polynomial model and regularization

A degree- d model includes an intercept, powers of individual inputs, and interactions between inputs. Each product $x_1^{a_1} \dots x_p^{a_p}$ satisfies $1 \leq a_1 + \dots + a_p \leq d$. For p inputs this gives $C(p+d, d)-1$ nonconstant terms. The search compares monomial and total-degree Legendre-product bases, both within the assignment's degree limits.

Polynomial columns are centred and scaled using the rows available for fitting. Regularization helps control large coefficients when many polynomial terms are included. Ridge minimizes $RSS + \alpha \|\beta\|_2^2$; Lasso minimizes $RSS/(2n) + \alpha \|\beta\|_1$. The intercept is not penalized, and the two α scales are different. Ridge uses an SVD solution. Lasso uses coordinate descent with optimality checks.

3. Validation procedure

For each dataset, a shuffled split with seed 197 sets aside 200 rows for holdout evaluation. The remaining 800 rows are used for model selection through five shuffled folds (seed 4197). Each fold fits 640 rows and validates on 160. Scaling uses only that fold's fitting rows. The model family, degree, basis and penalty are chosen by the lowest mean cross-validation MSE, without using the holdout scores.

The chosen model is fitted on the 800 development rows and evaluated on the 200 holdout rows. The same settings are then fitted on all 1,000 training rows to generate test predictions. Test inputs are used only for prediction, and their target values are unavailable.

Search settings

Search	Degrees	Penalty values
Ridge, both bases	var1: 1-10; var2: 1-20	13 powers of ten, 10^{-8} to 10^4
Lasso, monomial	var1: 3, 4, 5, 6; var2: 6, 8, 10, 12	0.1, 0.03, 0.01, 0.003

Lasso candidates with maximum fold duality gap above 0.001 are excluded. Final fitting uses coordinate tolerance 10^{-7} , compared with 10^{-5} during the search.

4. Data checks and correlation heatmaps

Figure 1 gives a first look at relationships in the training data. Correlation measures linear association, so it does not capture every relationship that a polynomial can model. All assigned inputs are retained, including those with small correlations with y .

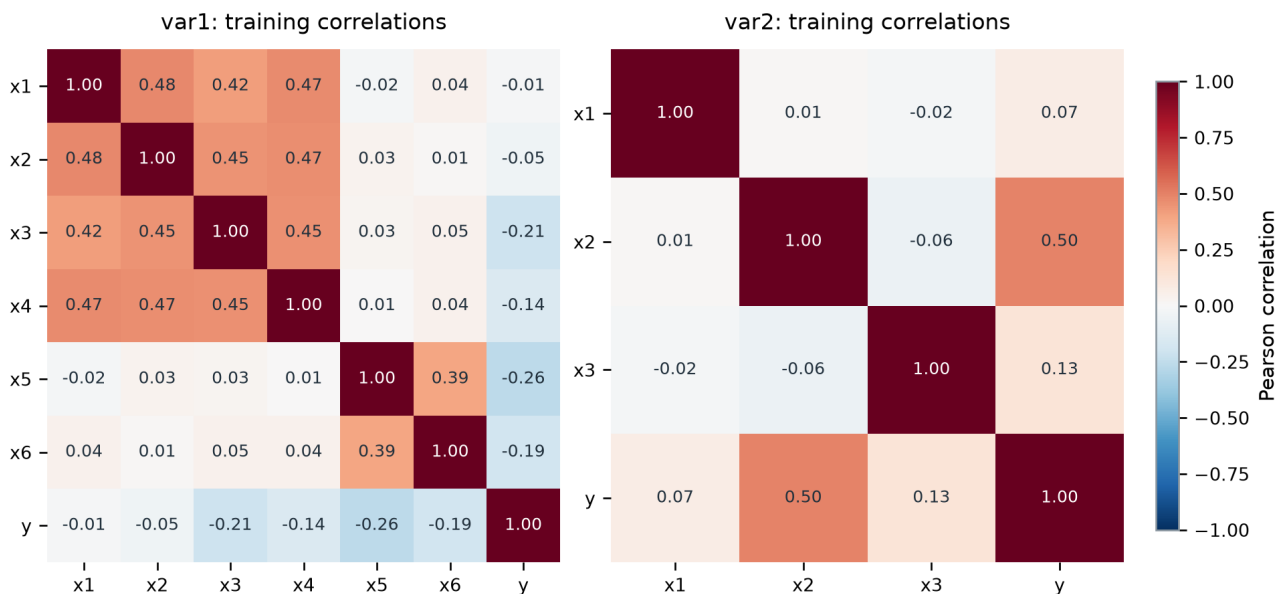


Figure 1. Pearson correlations from the 1,000 training rows for each problem. The y row and column describe association with the target. Only x columns are used as model inputs.

Training and test input coverage

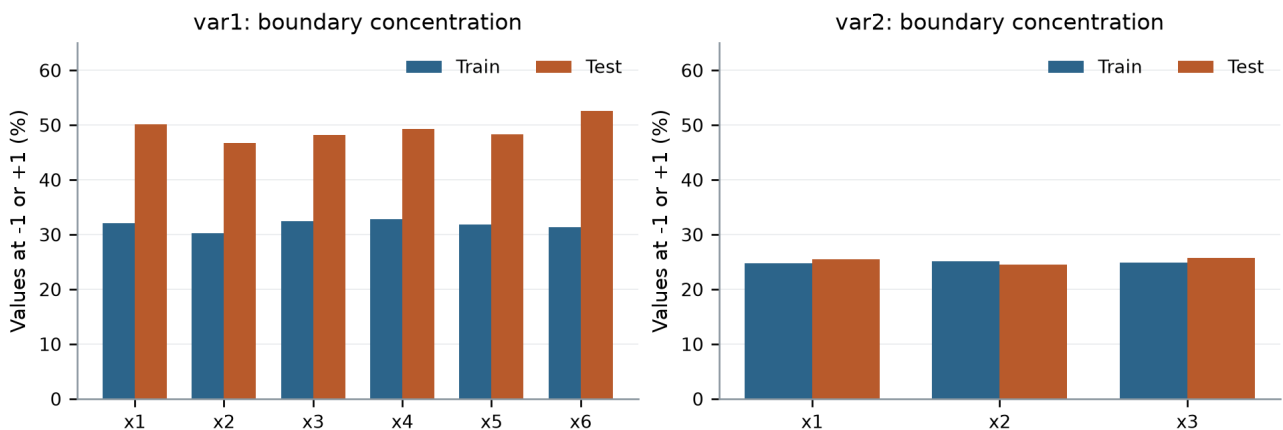


Figure 2. Share of train and test values equal to -1 or $+1$ for each input. Values are counted as supplied, without clipping or other changes.

The main train/test difference is in var1: 31.8% of training input entries and 49.2% of test entries are at -1 or $+1$. The proportions are closer for var2, at 24.9% and 25.3%. The extra boundary values in var1 test data mean that its hidden-test error may differ from the random holdout estimate.

Data check	var1	var2
Missing or non-finite values in any supplied file	0	0
Duplicate training feature rows	2	4
Duplicate test feature rows	2	6

Repeated input rows are not removed or averaged. Both prediction files follow the original test-row order.

5. Choosing the degree and penalty

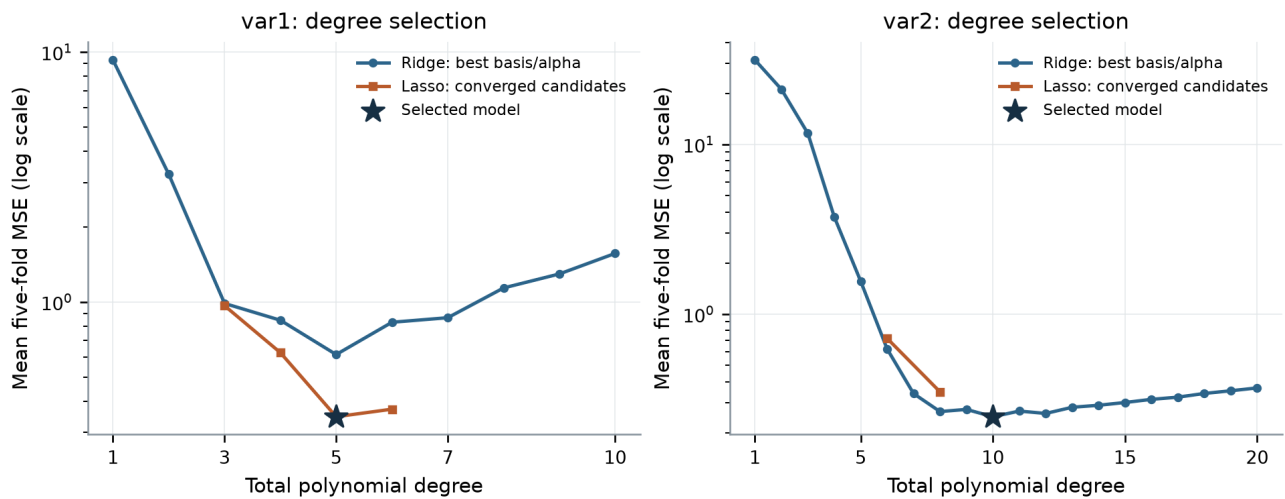


Figure 3. Mean five-fold MSE by total degree, on a logarithmic vertical scale. Each Ridge point uses its best basis and penalty. Each Lasso point uses its best converged penalty. Stars mark the selected models.

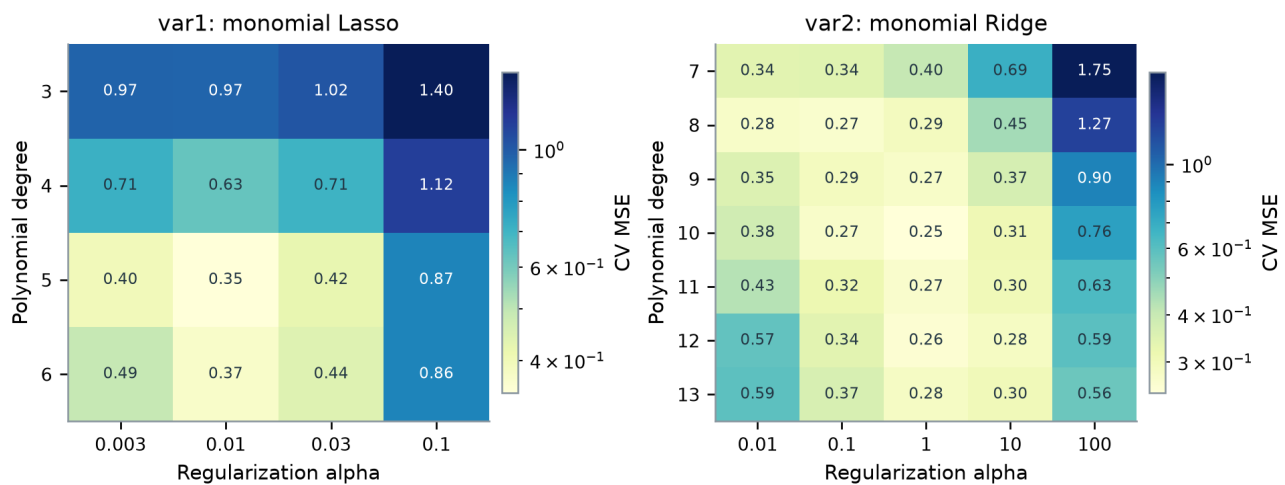


Figure 4. A closer view of the degree and penalty search. Numbers are mean CV MSE. Colours use a logarithmic scale separately in each panel. Excluded Lasso cells, if any, are grey. The panels show part of the original search.

Selected configurations and rationale

var1: Degree 5 Lasso, monomial basis, $\alpha = 0.01$, gave the lowest eligible CV MSE (0.34691; fold standard error 0.01301). The final model has 461 candidate polynomial terms and 119 nonzero coefficients.

var2: Degree 10 Ridge, monomial basis, $\alpha = 1$, gave the lowest eligible CV MSE (0.24842; fold standard error 0.02045). The final model has 285 candidate polynomial terms and 285 nonzero coefficients.

For var1, the best degree-4 Lasso MSE is 0.62623. Degree 5 reduces it to 0.34691, while degree 6 gives 0.37187. Degree 5 therefore improves the fit without adding terms that worsen validation error. Lasso keeps 119 of its 461 candidate coefficients nonzero. For var2, Ridge gives the better selection score and retains all 285 terms with coefficient shrinkage.

Degree 11 was included in the var2 Ridge search. Its best mean CV MSE was 0.26818, compared with 0.24842 for the selected degree 10 model. The limits 10 and 20 are upper bounds, not required degrees.

6. Holdout results and residual checks

The following scores and plots use the 200 holdout rows for each problem. Each model was fitted on the other 800 rows. This keeps the evaluation separate from the later fit on all training data.

Problem / model	Degree	CV MSE	Holdout MSE	Holdout R-squared
var1 / Lasso	5	0.346907	0.318792	0.970073
var2 / Ridge	10	0.248416	0.234471	0.996000

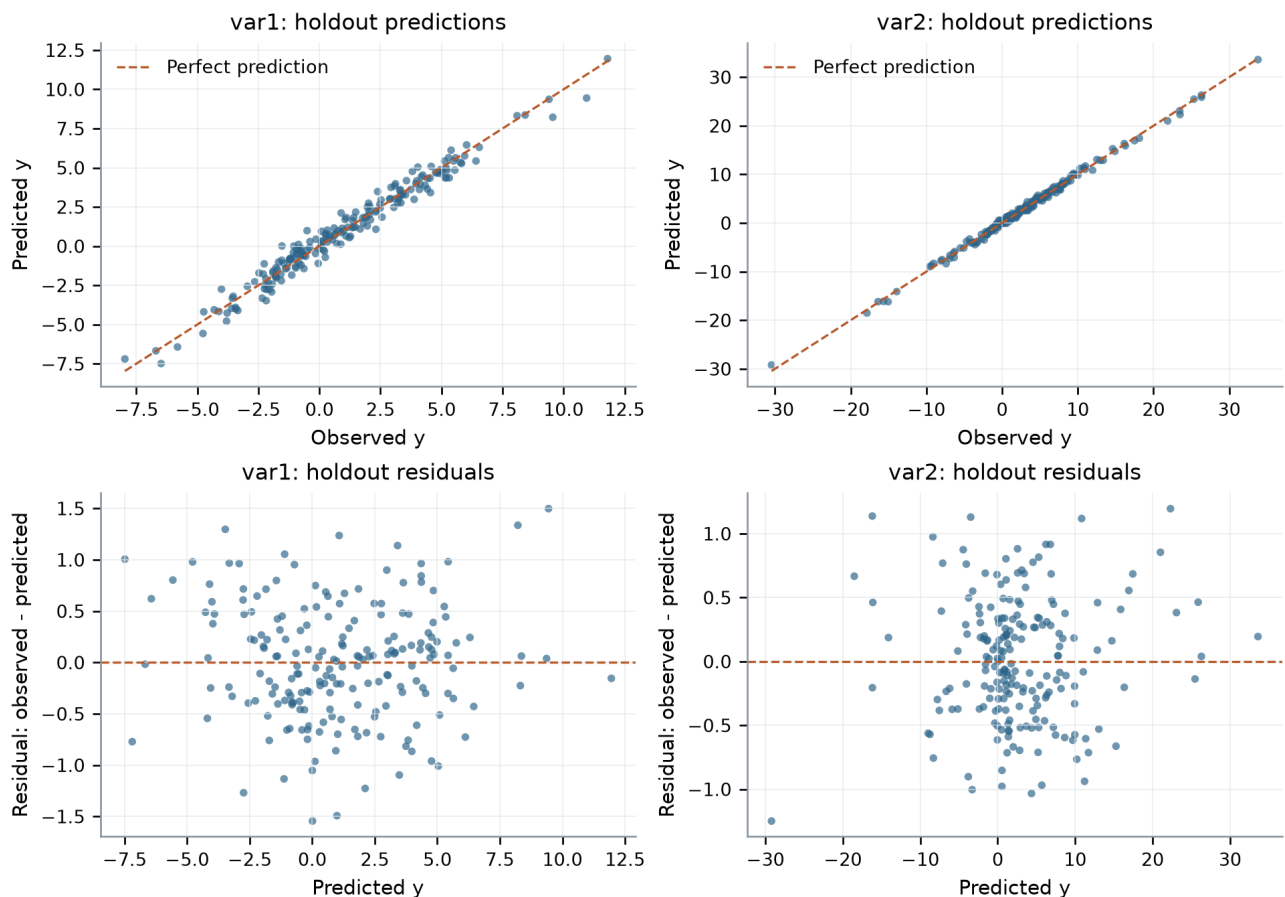


Figure 5. Top: observed versus predicted values, with the dashed line indicating perfect prediction. Bottom: residuals (observed minus predicted) versus predictions, with a zero reference line. Each panel contains 200 holdout observations; overlapping markers may hide repeated points.

$MSE = \text{mean}((y - \text{predicted } y)^2)$. $R\text{-squared} = 1 - \frac{\text{sum}((y - \text{predicted } y)^2)}{\text{sum}((y - \text{mean}(y))^2)}$. Both metrics are calculated from holdout predictions, not the hidden test targets.

For holdout observations with at least one boundary input, MSE is 0.31317 for var1 (181 rows) and 0.26998 for var2 (123 rows). Full-data training MSE is 0.24895 and 0.15640, respectively; those training values describe fit rather than generalization.

One var2 holdout coordinate also appears in the development rows. The split is random, not grouped by coordinate, so it mainly estimates performance for this sampling pattern. It does not measure performance in an entirely new spatial region. The finite holdout size and the earlier model search also add uncertainty.

7. Model-response heatmaps

Each final model is evaluated on a 101 by 101 grid over x_1 and x_2 , with the remaining inputs fixed at zero. These are two-dimensional slices through the fitted polynomial, not observed ground-truth maps or averages over the training distribution.

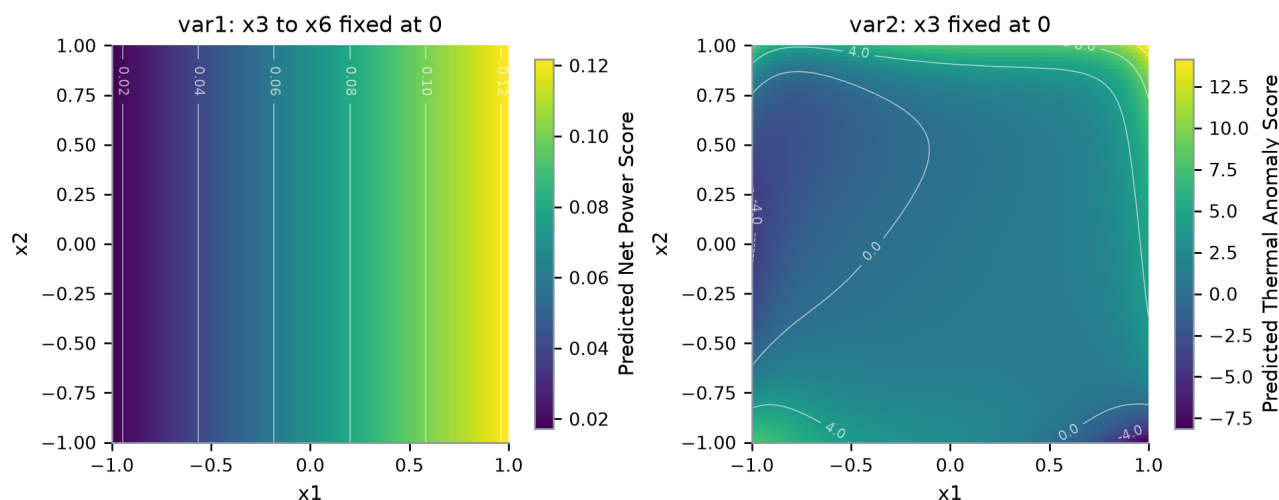


Figure 6. Predicted response slices, evaluated using the models fitted on all training rows. Left: Net Power Score with x_3 - $x_6 = 0$. Right: Thermal Anomaly Score at $x_3 = 0$. Contours join equal predicted scores. Colours have separate scales and are not comparable between the two problems.

The var1 map shows the response to the first two operating settings when the other controls are zero. The var2 map shows a horizontal reservoir slice at the supplied zero depth offset. Other fixed settings would give different maps. The submission predictions use all inputs from each actual test row, rather than these fixed-input grids.

8. Final predictions, limits and reproducibility

The chosen configurations are refitted on all 1,000 training rows. Each exported CSV contains exactly 1,000 finite predictions in the corresponding test-file order, with the single header `y` and no index column. The layout matches `sample_submission.csv`. Reloaded-model predictions and fresh full-training fits reproduce the exported values.

No test MSE or test R-squared can be calculated because test targets are hidden. The main limits of the evaluation are the var1 boundary shift, the size of the holdout sets and repeated coordinates. The holdout results support the selected models, but do not guarantee the same scores on the test files.

The repository includes the four datasets, sample CSV, training code, model-selection code and inference code. From the repository root, install `requirements.txt` and run:

```
python src/train.py --search-only
python src/search_sparse.py
python src/finalize.py --sample sample_submission.csv
python src/verify.py
```

Saved-model inference is available through `python src/predict.py --variant 1 (or 2)`. Use an empty results folder when changing the data or search configuration because the sparse search reuses completed results.

Separate course uploads	Contents
BT2024197_report.pdf	This five-page report, including the code repository link
BT2024197_pred_var1.csv	Net Power Score predictions
BT2024197_pred_var2.csv	Thermal Anomaly Score predictions

Sources: course assignment handout and submission instructions; the four supplied BT2024197 CSV files; `sample_submission.csv`. All figures are computed from those data, development-search records, holdout predictions or fitted model parameters.